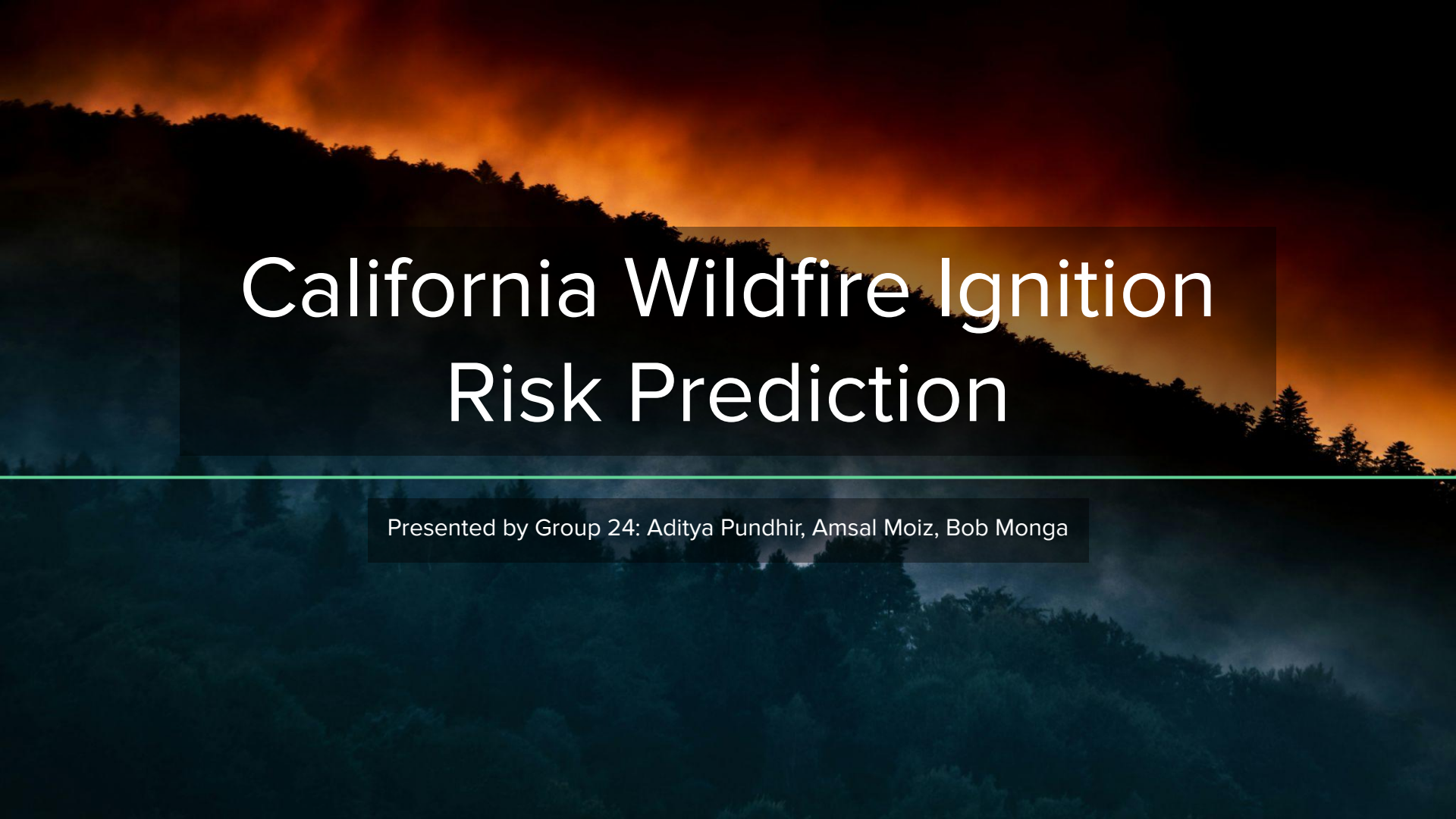


California Wildfire Ignition Risk Prediction

Presented by Group 24: Aditya Pundhir, Amsal Moiz, Bob Monga

Motivation: California's Escalating Wildfire Crisis

- Climate Change & Extended Fire Seasons
  - Rising temperatures and multi-year droughts drive longer fire seasons
  - High vapor pressure deficits (drier air) accelerate fuel drying and ignitions
- Socio-Economic & Ecological Impacts
  - Severe loss of life, displacement, and billions in structural/infrastructure damage
  - Secondary regional crises: toxic particulate air quality ($PM_{2.5}$), habitat loss, and post-fire flash floods



Core Issue: Shifting from Reactive to Predictive

Current Paradigm (Reactive)

- **Post-Ignition Focus**: Deploys resources after smoke or active burns appear
- **Active Simulators**: Models require a confirmed fire to simulate spread
- **Coarse Granularity**: County-wide warnings lack local precision

VS

Our Approach (Predictive)

- **Pre-Ignition Modeling**: Forecasts ignition likelihood *before* fires ignite
- **Localized Resolution**: Evaluates risk across a uniform 5km X 5 km grid
- **Multimodal Fusion**: Merges daily weather, live fuel moisture, and terrain

Multimodal Data Fusion: Mapping Drivers to Datasets

Topography (USGS DEM)	Weather Dynamics (ECMWF ERA5)	Fuel Moisture (MODIS)	Ignitions (FPA-FOD)
Role: 5km static terrain anchor **Key Variables**: - **elevation**: microclimates - **slope**: uphill flame spread - **aspect**: heading & sun exposure	**Role**: Daily fire weather dynamics **Key Variables**: - **t2m & rh2m**: heat & fuel drying - **wind_speed_10m**: oxygen & spread - **tp**: precipitation suppression	**Role**: Vegetation drought stress **Key Variables**: - **ndvi**: Live canopy biomass density - **evi**: High-density vegetation health	**Role**: Ground-truth target labels **Key Variables**: - **valid_time**, **FIRE_YEAR**, **LAT**, **LON** - **fire_occurred (0, 1)**: Binary ignition event classification

Pipeline Architecture & Spatial Harmonization

The Spatial Challenge

Resolution Mismatch: ERA5 NetCDF grids and MODIS rasters used different native coordinate systems and cell sizes relative to our 5km master grid.

Join Breakdown: Standard reverse spatial joins collapsed coverage to only 0.85% of California cells while multiplying duplicate records.

Memory Bottleneck: Inefficient merges threatened out-of-memory crashes on multi-year temporal panels (~82M rows).

The Vectorized Validation & Fix

Vectorized Interpolation: Replaced spatial joins with `xarray.interp(method="nearest")` using pre-indexed grid centroids, guaranteeing 100% spatial coverage.

Deduplication Engine: Applied `.groupby(['cell_id', 'valid_time']).mean()` aggregation to enforce exact 1-to-1 panel grain consistency.

Optimized Storage: Serialized into columnar Apache Parquet, delivering fast multi-threaded I/O and zero data leakage across splits.

Training Setup: Predicting Tomorrow's Ignitions

2022 — TRAIN

554,224

sampled observations

3,608 fire cases

›

2023 — VALIDATE

27.66M

full observations

3,668 fire cases

›

2024 — FINAL TEST

27.73M

full observations

3,843 fire cases

PRIOR-DAY ENVIRONMENTAL INPUTS → NEXT-DAY FIRE LABEL (0 / 1)

Extreme imbalance • All training fires kept • 2% negatives sampled with compensating weights

Three Models, Same Environmental Inputs

WEATHER
4 variables

VEGETATION
2 indices

TOPOGRAPHY
3 variables

↓ same nine inputs fed into each model ↓

LOGISTIC REGRESSION

Linear baseline

Simple, interpretable

scaled + imputed

RANDOM FOREST

Bagged decision trees

Captures nonlinear effects

100 trees · max depth 12

XGBOOST

Boosted decision trees

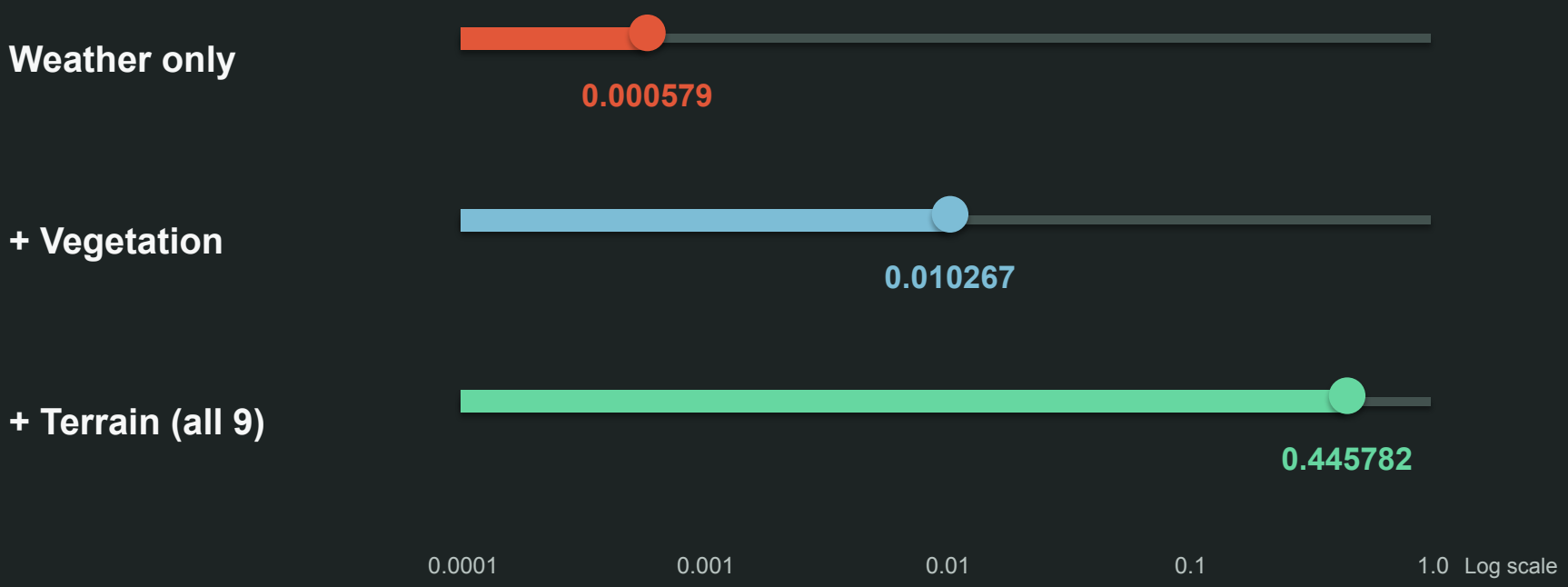
Learns sequential corrections

300 trees · max depth 5

Train on 2022 • choose thresholds on 2023 • evaluate once on 2024

Feature Ablation: Which Inputs Actually Help?

RANDOM FOREST • 2024 TEST • AVERAGE PRECISION (PR-AUC)



KEY INSIGHT Terrain adds a large signal

How Did We Evaluate Our Models?

Precision

- Measured how many predicted ignitions were correct
- Helps identify false alarm

Recall

- Measures how many actual ignitions were detected
- Helps us understand how many fires the model missed

F1-Score

- Balances precision and recall
- Helps compare overall detection performance

ROC-AUC

- Measures how well a model distinguishes between ignition and non-ignition cases

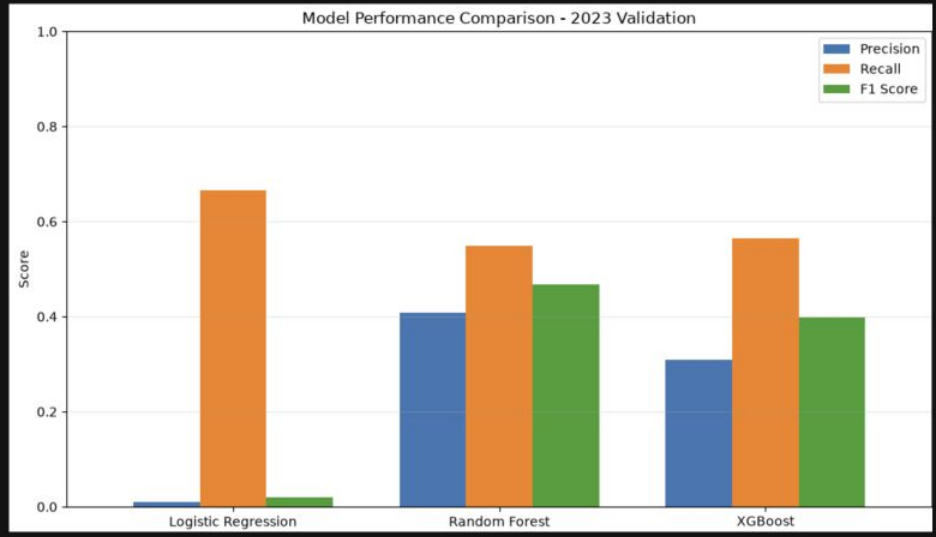
Average Precision

- Summarizes precision and recall across thresholds

Accuracy alone is not sufficient because wildfire ignition events are extremely rare compared to non-ignition events

Model Performance-2023 Validation

Model	Precision	Recall	F1-Score
Log Reg	0.009	0.663	0.0196
R.Forest	0.4073	0.5499	0.4680
XBoost	0.3088	0.5641	0.3991

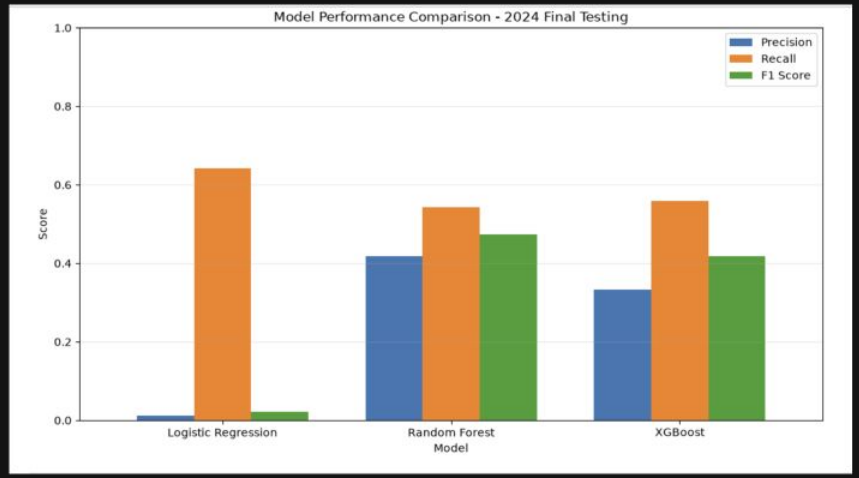


Key Observations:

- Logistic Regression: Highest recall, but low precision
- Random Forest: Best balance between precision and recall
- Xboost: Intermediate performance between the other models

Model Performance-2024 Final Testing

Model	Precision	Recall	F1-Score
Log Reg	0.0112	0.6419	0.0220
R.Forest	0.4189	0.5438	0.4733
XBoost	0.3329	0.5587	0.4172



Key Observations:

- Logistic Regression continued to achieve the highest recall
- Random Forest maintained the highest precision and F1-Score
- Xboost remained competitive but did not outperform Random Forest overall

Random Forest maintained the strongest overall balance on the final test 2024

Which Model Perform the best?

Logistic Regression

- Highest recall
- Detect the largest proportion of ignitions
- Generated many false alarms

Random Forest

- Highest precision and F1-Score
- Strong ROC-AUC and average precision
- Provided the best overall balance

XBoost

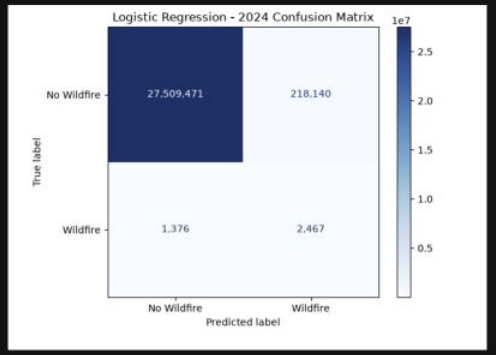
- Competitive precision and recall
- Better balance than Logistic Regression
- Did not surpass Random Forest

What about ROC-AUC and Average Precision?

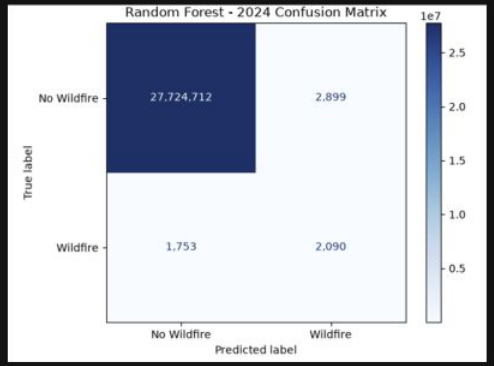
Model	2024 ROC-AUC	2024 AP
Logistic Regression	0.9657	0.0120
Random Forest	0.9910	0.4458
XBoost	0.9458	0.2704

Confusion Matrix Analysis

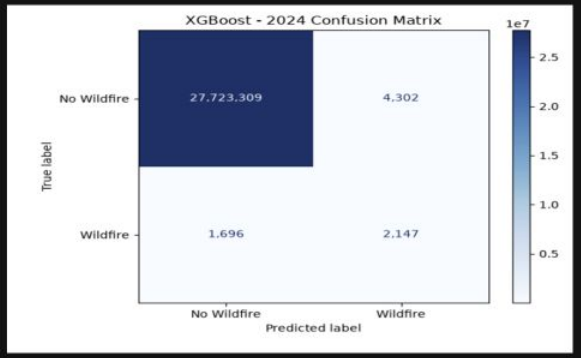
Logistic Regression



Random Forest



XBoost



Key Takeaways:

- All models correctly classified most non-ignition cases(class0)
- Random Forest had the fewest false positives(903)

V Logistic Regression detected the most ignitions(highest recall, 338), butwith many more false alarms

Final Results and Model Recommendation

Random Forest was the best overall-performing model

Best Overall Balance

Achieved the strongest balance between detecting wildfire ignitions and limiting false alarms

Consistent Performance

Maintained strong results across both the 2023 validation and 2024 testing datasets

Important Trade-off

Although Logistic Regression achieved the highest recall, its very low precision resulted in many false alarms